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Started	Wednesday, 21 October 2026, 9:52 PM
Completed	Wednesday, 21 October 2026, 9:56 PM
Duration	3 mins 38 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted
b. The search continues from the beginning
c. The search stops
d. The search continues in the left or right sublist ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Linear search
c. Sequential search
d. Binary search ✓

Question **3**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The search is considered unsuccessful ✓
c. The last element is returned
d. The first element is returned

Question **4**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Hash search
- c. Binary search
- d. Both (1) & (3)

Question **5**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search continues in the left sublist
- b. The search ends successfully ✓
- c. The list is sorted
- d. The search continues in the right sublist

Question **6**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. n ✓
- b. log n
- c. n/2
- d. 1

Question **7**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search
b. Hash search ✗
c. Linear search
d. Interpolation search

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list
b. In the left sublist ✓
c. In the entire list
d. In the right sublist

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted
b. Number values should only be present

Number values should only be present

c. There should be direct access to the middle element in any sublist
d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **10**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly
b. The search stops
c. The higher middle element is chosen as the middle element
d. The lower middle element is chosen as the middle element ✗

Question **11**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(n log n)
c. O(log n) ✓
d. O👎

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search requires the list to be sorted.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search divides the list into halves.

Question **13**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list ✓
b. Divide the list into two equal parts
c. Sort the list
d. Compare the target element with the first element in the list

Question **14**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is small and sorted
c. When the list is small and unsorted ✓
d. When the list is large and sorted

Question **15**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By comparing each element in the list sequentially ✓
b. By using a hash function
c. By dividing the list into halves
d. By sorting the list first